Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Shackelford, Linda M.D.		
Project Title:	Prevention of Bed Rest Osteoporosis: Resistive Exercise		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	283-483-7100
Organization Name:	NASA - Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-OLMSA-01
Start Date:	05/31/1997	**End Date:**	10/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Note - end date changed from 10/31/2003		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Using 17 weeks horizontal bed rest as an analogue for space flight, a resistive exercise countermeasure was tested for efficacy in preventing bone and muscle loss and associated metabolic changes. Overall results were encouraging. Urinary calcium excretion decreased in exercisers. Bone formation markers and resorption markers increased compared to baseline. Lumbar spine density increased in exercisers and remained unchanged in controls. Calcaneus did not lose bone in exercisers, while bed rest controls showed losses equal to space flight. The femoral trochanter was unchanged in either group. The groups as a whole did not differ in femoral trochanter losses. However, the manner in which exercises were performed appeared to have a significant influence upon greater trochanter bone mineral density (BMD) changes. Isotonic strength increased in exercisers and decreased in the lower extremities in bed rest controls. Exercise loads for 5 rep max exceeded 600 pounds in some cases. The results of this study contributed significantly to ISS exercise equipment and protocol requirements development.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Proof that heavy loads without impact can maintain calcaneus BMD has advanced underatanding of mechanical factors required for exercise protocols to prevent or treat osteoporosis. This will be of benefit to persons desiring an exercise program to benefit bone but have conditions that prohibit jumping and jogging. Identification of specific exercises that influence specific bone regions may lead to targeted exercise programs to maximize benefits to osteopenic and osteoporotic patients.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)